

Daniel Lofeodo

Montreal, Canada
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Education

- 2024-08 – 2026-05
- M.Sc. Computer Science (Machine Learning)**
MILA, Université de Montréal, Montreal, Canada
 - 4.2/4.3 GPA
- 2026-05 – 2026-06
- Certified Generative AI Developer – Professional** (*Cloud ML & MLOps*)
Certified Machine Learning Engineer – Associate (*Cloud AI Specialization*)
Amazon Web Service, Online
- 2017-08 – 2021-05
- B.Eng. Mechanical Engineering**
Concordia University, Montreal, Canada
 - Dean's List (2021)

Spotlight Project

- 2026-05 – 2026-06
- Cloud-Native Agentic AI Pipeline — Automated AI-Industry Newsletter**
Built an end-to-end cloud-native agentic AI app: an autonomous LLM (Claude) pipeline delivering a weekly AI-industry newsletter, with Python microservices on GCP, FastAPI, Docker, and Firebase.

Experience

- 2025-03 – 2026-01
- Machine Learning Researcher**
Hydro-Québec, Montreal, Canada
Artificial intelligence agents for hydro-dam operation during 1 in 10,000-year floods.
 - Built an end-to-end ML pipeline (data ingestion → feature engineering → training → evaluation) for hydraulic system modeling.
 - Developed a constrained reinforcement learning framework to evaluate hydro-dam behavior under extreme, out-of-distribution (10,000-year) flood scenarios.
 - Designed a layered safety mechanism (hard constraints + dynamic reward shaping) to enforce safety-critical operating limits.
 - Benchmarked state-of-the-art supervised models for multi-step time-series forecasting of hydrological variables.
 - Synthesized research in deep learning for time-series to guide modeling strategy and model selection.
- 2021-06 – 2024-08
- Software Engineer**
Creaform, R&D, Montreal, Canada
High precision aerospace-grade 3D scanners and software for part inspection and reverse engineering.
 - Developed core features for aerospace-grade 3D inspection software, including geometric tolerancing (circularity, cylindricity, flatness) for validating scanned parts against engineering specifications.
 - Built a new industrial pipe inspection product by repurposing internal components, contributing to a high-revenue product line.
 - Led UI/UX development for Creaform.OS, improving performance through profiling and delivering customizable interfaces.
 - Contributed to large-scale C++ codebases in an Agile environment with CI/CD, code reviews, and continuous refactoring.
- 2020-05 – 2020-09
- Research Assistant**
National Science and Engineering Research Council, Montreal, Canada
Autonomous robot for weed removal on private lawns.
 - Developed an end-to-end computer vision pipeline for an autonomous weeding robot, including CNN-based weed detection, automated image preprocessing, and integration with mechatronic actuation.

Technical Skills

- Programming Languages:**
Python, C++, C#, XAML
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- SQL, Java, JavaScript, TypeScript, CSS, HTML, dart
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- XML, Lua
●●●○○
- AI Frameworks:**
PyTorch, AutoML, scikit-learn, xarray, XGBoost, Pandas, Numpy, stable-baselines3, gymnasium, etc.
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- UI Frameworks:**
Flutter, Three.js, WPF, WinForms, matplotlib, Seaborn
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- Streamlit, Flask
●●●●○
- Cloud and DevOps:**
Azure, GCP, AWS, GitHub, Databricks, Docker
●●●●●
- Environments:**
Windows, WSL, Databricks
●●●●●
- Linux
●●●●○
- Project management:**
Asana, Jira, Teams
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Soft Skills

- Competent
- Reliable
- Consistent
- Punctual
- Ability to receive feedback with humility
- Positive

Languages

- English (Native), French (Native)
●●●●●
- Italian (Advanced - C1)
●●●○○

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